

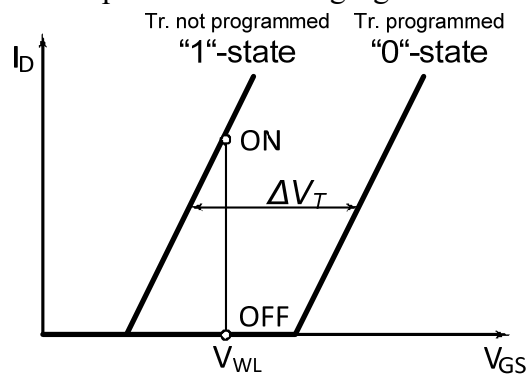
Chapter A Combinatorial Circuit Design

1. A CMOS-inverter is sized as a symmetric unit-inverter with the following parameters: $L_{NMOS} = L_{PMOS} = L_{\min}$, $W_{NMOS} = W_{\min}$, $W_{PMOS} = 2 \cdot W_{\min}$
This CMOS-inverter is driving a symmetric unit-inverter of the same type (load)!
 - a) Draw the transistor schematic of this circuit! Indicate the transistor width of each transistor in this schematic.
 - b) Assuming a rising edge on the input of the first inverter. Draw an equivalent RC-circuit by using the RC-model of a MOS-transistor! Assume the diffusion capacitances are equal to the gate-capacitance: $C_g = C_{sb} = C_{db} = C$
 - c) Give a formula for the propagation delay t_{pdHL} of the first inverter as a function of RC-delays, assuming a rising edge on the input!
2.
 - a) Draw the transistor schematic of a transmission-gate!
 - b) Describe the main advantage of using a transmission-gate instead of a NMOS-pass-transistor! Distinguish two cases: passing a '1' and passing a '0' from the input to the output.
3. Draw the transistor schematic of a static CMOS-gate implementing the following function: $Y = \overline{A \cdot (B + C)}$

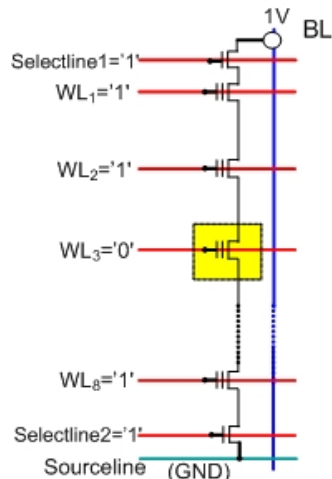
Chapter B Memory Circuit Design

4. Name two different categories of RAMs regarding its behavior on the power supply!
5. Draw a block diagram of an array-structured memory architecture!
6. Name two advantages of hierarchical memory architecture!
7. Draw the circuit schematic of a 1-transistor DRAM cell!
8. Why DRAMs need to get refreshed?
9. Describe the DRAM write operation: assume a logical '1' needs to get written into a cell currently storing a logical '0'!
10. Assume a logical '1' is stored into a DRAM-cell. Describe the 3-steps of the DRAM read operation! Draw also a voltage vs. time diagram for the bitline signal during read operation!
11. What is the advantage of using a folded bitline architecture instead of a open bitline architecture for a DRAM ?
12. What is the main characteristic of a DDR-SDRAM regarding its I/O data transfer?
13. The burst length of a SDRAM is programmable. What is the meaning of burst length and how can the burst length be programmed?

14. DDR3 SDRAM uses an 8n Pre-fetch vs. an 4n Pre-fetch of DDR2. What is the advantage of 8n Pre-fetch?
15. DDR3 SDRAM uses 1.5V power supply vs. 1.8V power supply of DDR2. What is the advantage of reduced power supply voltage?
16. eDRAM is considered as a replacement for which memory type? For which application?
17. Draw the circuit schematic of a 6-transistor SRAM-cell!
18. Why SRAM-cells do not need to get refreshed?
19. Assuming a logical '0' is stored into a SRAM-cell. Describe the 3-steps of the read operation! Draw also a voltage vs. time diagram for the most important signals and nodes of the SRAM-cell during the read operation!
20. Draw a cross-section of a floating-gate transistor (FAMOS)!
Explain the following figure!



21. Explain the Read-Operation of a NAND Flash memory!



22. SSD is replacing HDD. Name two advantages/disadvantages of SSD over HDD!
23. Summarize possible advantages of PRAM over NAND- and NOR-flash! How does PRAM work? Explain the two stable states of a PRAM device!

Chapter C&D**Design Methodology & Design Verification**

24. We distinguish the array-based implementation approaches from the cell-based implementation approaches. Nowadays FPGAs are a popular choice for an array-based implementation approach.
- Name three main advantage of choosing a FPGA implementation instead of an ASIC implementation (standard-cell design)!
 - Draw a figure of a simplified FPGA architecture!
- 25.
- Give three reasons from an application point of view why SoC is the recent trend!
 - Draw a figure of a typical SoC architecture!
 - To increase design productivity SoC design uses the “core based design approach”. Give a definition of the term “core”! Give two examples!
 - Name five main SoC design challenges!
26. Name two approaches how FPGAs are used for design verification!

Chapter E**IC Fabrication**

27. The following figure shows the cross-section of a wafer while manufacturing the n-well of a CMOS-process assuming a p-substrate as the starting material. Describe the manufacturing steps (b)-(h) to manufacture the n-well. Give a short description of each step on the right hand side of the figure!

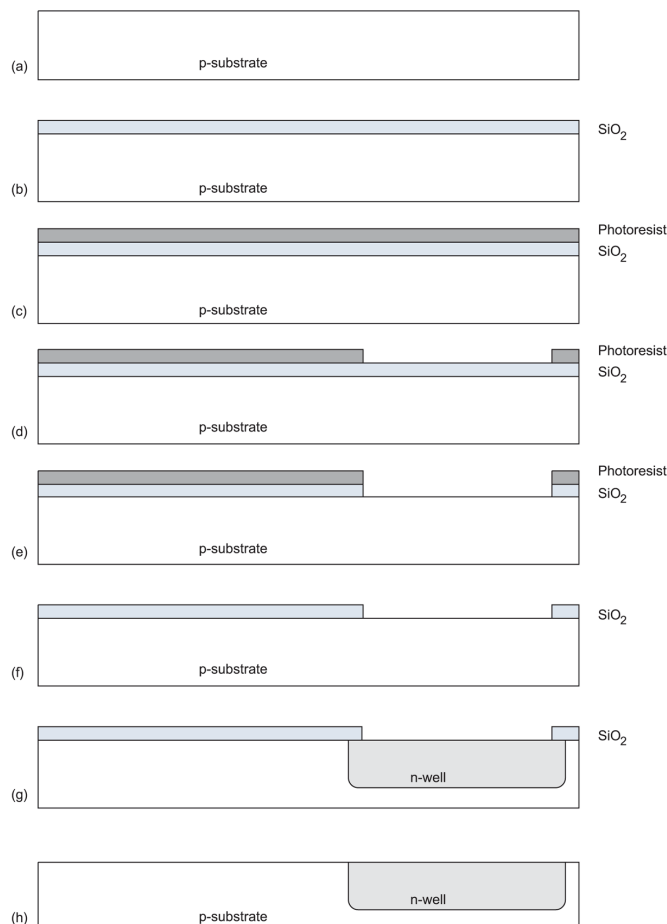
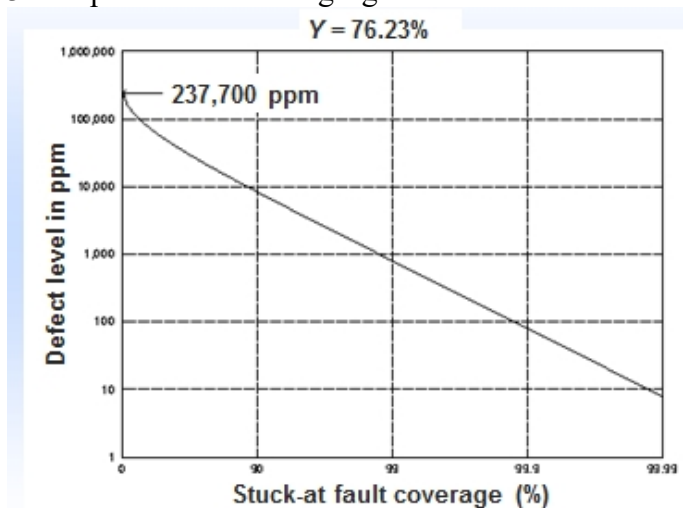


FIG 1.36 Cross-sections while manufacturing the n-well

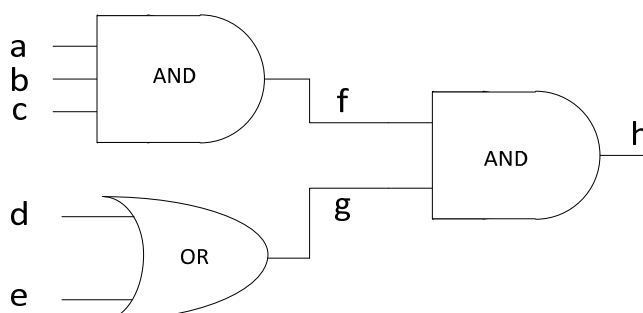
28. A simple n-well CMOS-process should get manufactured. The starting material is a p-substrate, called a “wafer”. A mask set of at least 6 masks are necessary to define the correct position of layers. Name the six different layers for which the masks are designed in the correct order of the manufacturing process!
29. Name one advantage of EUV lithography!
30. a) Define the term “yield of integrated circuit wafer manufacturing”!
b) Give a formula for the manufacturing cost of a fully functional chip as a function of wafer cost, number of chips per wafer and manufacturing yield of the chip!

Chapter F IC Testing

31. Define the term “verification” and the term “test”!
32. Explain the following figure!

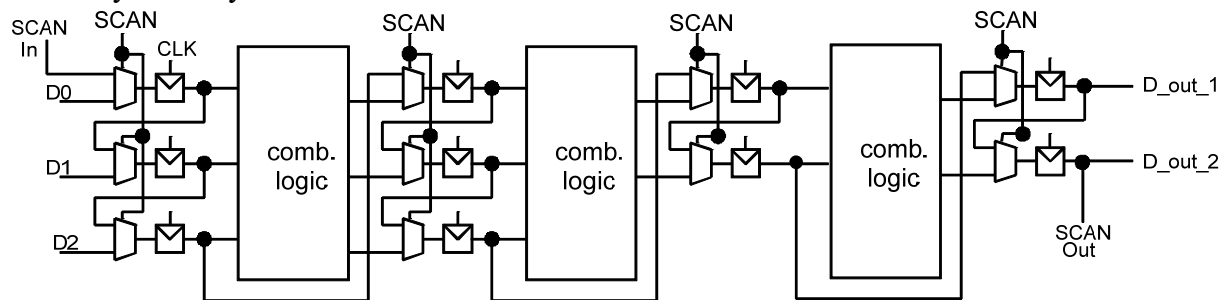


33. Define the following terms: fault coverage, defect level
34. The stuck-at-fault model is the most important model to describe physical defects. To generate test vectors which detect a stuck-at fault at a target node a systematic method is used. Describe the algorithm to detect a stuck-at fault on a target node!
35. The following circuit comes with five primary input and one primary output. Determine all test vectors which can detect a possible stuck-at-0 fault on node g! Determine all test vectors which can detect a possible stuck-at-1 fault on node g!



36. Explain the steps of testing the sequential circuit with scan path shown in the figure below!

How many bits one test vector (pattern) needs? How many bits one test result have? How many clock cycles the test takes for one test vector?

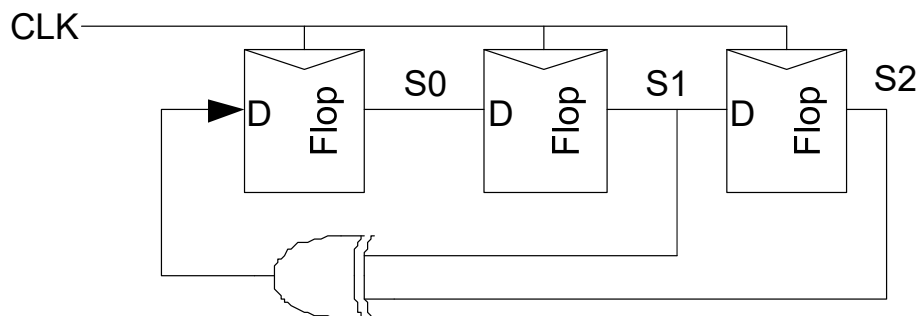


37. What is main advantage of using scan design?

38. What is the design overhead implementing a scan path?

39. Explain the term BIST! Draw a block diagram of a general BIST structure!

40. The following circuit shows a LFSR working as a PRSG. Determine the sequence of test vectors which are generated, assuming an initial output $S_0=1, S_1=0, S_2=0$.



41. If the all-0s test is required, the 3-bit LFSR shown above can be modified adding a NOR-gate. Modify the schematic of the 3-bit LFSR and determine the state sequence, assuming an initial output $S_0=S_1=S_2=0$!